Introduction to Module 2: Randomization

Design and Analysis of Algorithms

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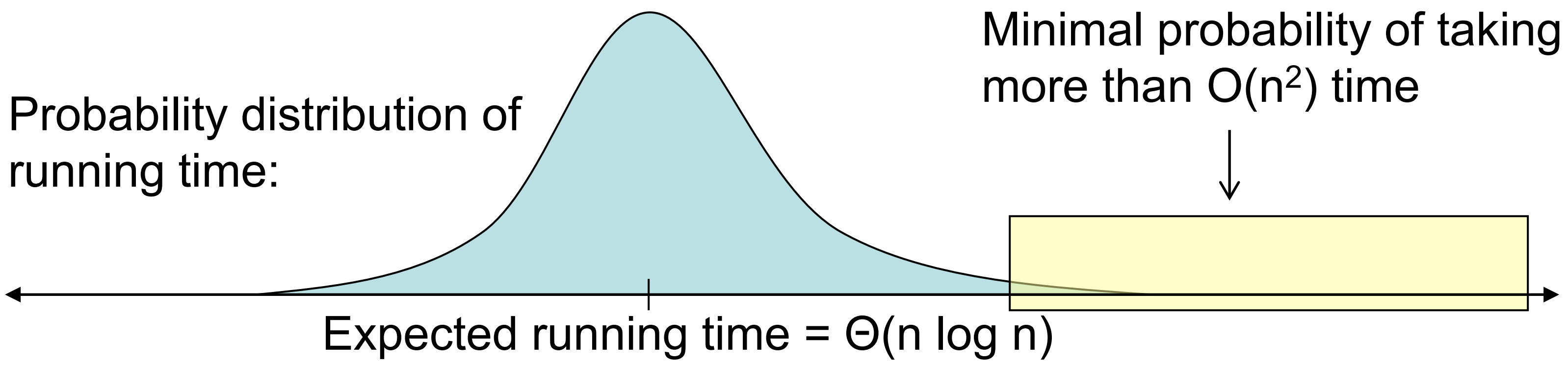
Module Goals

- Learn common techniques for randomized algorithm analysis and average-case analysis.
- Introduce and analyze examples of several prominent randomized algorithms.
- Understand relevant ideas from probability theory (e.g., expected value, tail bounds, linearity of expectation, union bounds)

Types of Randomized Algorithms and Analyses

- With a Monte Carlo algorithm, we typically want to prove that the output is correct with high probability.
- With a Las Vegas algorithm, we typically want to show that a certain running time bound holds in expectation or with high probability.
  - Same goals in average case analysis of non-random algorithms.



Module Structure and Notes

- Exercises for practice
- Coding: Priority queues and randomly-mergeable binary heaps
- Enrichment lecture (optional): Connections with information theory (e.g., entropy and its relatives, mutual information, compression, …)

- We're studying randomization early since it's useful in a <u>very</u> wide range of algorithmic situations.
- Perhaps a bit more analytical than other modules.
- Lots of cool and elegant ideas in this domain!